



Abhinav Shukla

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B.Tech in Computer Science and Engineering

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EDUCATION

•Indian Institute of Information Technology Bhagalpur

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – Present

CGPA: 8.89/10.00 (*Absolute Grading)

•DAV Public School Robertsganj, Sonbhadra

Senior Secondary (Class XII), CBSE

Apr 2021 – Mar 2022

Percentage: 84%

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, Python, Kotlin, Java

Android Development: XML, Jetpack Compose, Retrofit, Coroutines, Firebase, Material3

Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, Keras, MediaPipe, Streamlit, Matplotlib

Database Systems: MySQL, SQLite, MongoDB

Cloud/DevOps: Docker, Linux, Firebase Hosting, REST APIs

Version Control: Git, GitHub

Soft Skills: Debugging, Mentorship, Collaboration, Communication

Related Courseworks: Data Communication, Computer Networks, Data Structure and Algorithms

PROJECTS

•Rockfall Detection & Prediction System

Sep 2025 – Oct 2025

YOLOv8, XGBoost, Random Forest, FastAPI



- Designed an AI-powered system for real-time rockfall detection using YOLOv8 and ensemble ML models (78.8% accuracy).
- Integrated DEM terrain analysis, environmental sensors, and weather data for predictive risk assessment.

•Smart AI Traffic Automation

Apr 2025 – May 2025

Q-learning, YOLOv8, OpenCV, TensorFlow, Cryptography



- Led development of an AI-driven traffic system using Q-learning, reducing congestion by 30%.
- Implemented real-time YOLOv8 detection with 95% accuracy.
- Secured 10,000+ data transactions between sensors and control nodes using cryptographic validation.

•Secure VPN Tunnel

Jan 2025 – Mar 2025

Python, AES-256-GCM, PBKDF2, Sockets



- Engineered a secure VPN prototype using AES-256-GCM and PBKDF2 for encrypted end-to-end communication.
- Implemented multi-client connections with detailed throughput logging, improving stability and reliability

•Password Manager App

Oct 2024 – Jan 2025

Kotlin, Jetpack Compose, Room, MVVM, AES-256-GCM



- Developed a secure password manager with AES-256-GCM encryption and unique keys per installation.
- Added CRUD operations, real-time search, and a customizable password generator.
- Applied Clean Architecture (MVVM + Repository) to ensure scalability & maintainability.

POSITIONS OF RESPONSIBILITY

•Technical Mentor, Adhyaay, IIIT Bhagalpur

Aug 2024 – Aug 2025

- Mentored 20+ students in Android Development & DSA.
- Guided project teams on Android & ML, fostering hands-on learning.

•College Ambassador, Kshitij, IIT Kharagpur

Jul 2024 – Jan 2025

- Promoted IIT KGP's annual fest to over 200 students.
- Acted as a liaison between organizers and IIIT Bhagalpur.

ACHIEVEMENTS

•Codeforces– Pupil, Max Rating: 1244 (codeforces.com/tron_)

•CodeChef– 2 star, Max Rating: 1495 (codechef.com/tron_00)

•LeetCode and GfG– Max Rating:1458 on LeetCode, Solved 500+ DSA problems combined ([Leetcode/GfG](https://leetcode.com/))

•Smart India Hackathon– Secured 4th place among 50+ participating teams (College Round)

•Amazon ML Challenge 2025– Secured All India rank of 985 with a SMAPE score of 50.04.